



Modelling Financial Markets

Chapter 4: Portfolio Optimisation



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample

Acronyms used in this material (2/2)

- POET: Principal Orthogonal compleMent Thresholding
- US: United States

Outline

Motivation

Mean–Variance Optimisation

The Limits of Markowitz

Covariance Shrinkage

Black–Litterman

Risk Parity and Hierarchical Risk Parity

Out-of-Sample Evaluation

A Portfolio of BVB Blue Chips

Case Study: The Implementable Efficient Frontier

AI for Scientific Discovery

Conclusions

References

Portfolio Optimisation: Easy to Write, Hard to Estimate

- A **portfolio** is a weight vector $w = (w_1, \dots, w_N)^\top$ with $w^\top \mathbf{1} = 1$; optimisation chooses w to maximise a criterion under constraints
- The inputs are **estimated**: N means and $N(N+1)/2$ variances and covariances
 - ▶ 16 assets: $16 + 136 = 152$ parameters from a 60-month window, i.e. 960 observations
- Uses: strategic allocation (pension funds), home bias of private investors, the covariance matrix of risk management (Chapters 6–7)
- Key message: judge every rule **out of sample**, after costs, with a test of the difference
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), Part I (portfolio theory)

The Central Tension

In theory

- ▣ Known means and covariances
- ▣ The mean–variance portfolio is optimal
- ▣ More assets always help

In practice

- ▣ Parameters are estimated with error
- ▣ The optimiser loads on the largest errors
- ▣ More assets, more parameters, more noise

- ▣ This chapter: better inputs (shrinkage, Black–Litterman), rules that ignore means (risk parity, HRP) and a fair out-of-sample comparison

Data Used in This Chapter

Universe	Assets	Period
Sectors	9 US sector ETFs (XLB ... XLY)	1999–2026
Multi-asset	SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD	2007–2026
Combined	9 sectors + 7 multi-asset ETFs (no SPY)	2007–2026
Romania	8 BVB blue chips, BET, BET-TR	2014–2026

- Monthly returns from month-end prices on common trading days; US data end in July 2026, BVB data in August 2026
- Risk-free rate: one-month Treasury bill, [Kenneth French Data Library](#)

The Assets

- ▣ Equities: SPY (S&P 500), EFA (developed markets outside North America), EEM (emerging markets)
- ▣ US Treasuries: TLT (maturities above 20 years), IEF (7–10 years)
- ▣ Credit: LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds)
- ▣ Gold: GLD; sectors: XLK technology, XLU utilities, XLP consumer staples, . . .
- ▣ BVB: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET is a price index, BET-TR reinvests dividends

Prerequisites Recap: Notation and Two Assets

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$; portfolio mean $w^\top \mu$, variance $w^\top \Sigma w$
- **Sharpe ratio** (Sharpe (1966)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$; monthly figures annualised with $\times 12$ (mean) and $\times \sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t (W_t / \max_{s \leq t} W_s - 1)$, the largest fall of wealth W_t from a previous peak
- Two assets, correlation ρ : $\sigma_p^2 = w^2 \sigma_1^2 + (1-w)^2 \sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$, minimised at $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026: $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against 14.7% and 13.5% (time-varying correlation: Chapter 6)

The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean m ; its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Utility form, risk aversion $\gamma > 0$: $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$ s.t. $w^\top \mathbf{1} = 1$; as γ varies it traces exactly the efficient branch
- ▣ Inputs: N means, N variances, $N(N-1)/2$ covariances
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Deriving the GMV and Tangency Portfolios for N Assets

- GMV: $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ s.t. $\mathbf{w}^\top \mathbf{1} = 1$; Lagrangian $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$
 - ▶ First-order condition $2\Sigma \mathbf{w} = \lambda \mathbf{1}$, so $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
 - ▶ $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1}/A$ and $\sigma_{\text{GMV}}^2 = \mathbf{w}_{\text{GMV}}^\top \Sigma \mathbf{w}_{\text{GMV}} = 1/A$
- $N = 2$: $\Sigma^{-1} \propto \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}$ returns $\mathbf{w}^*$ of the recap slide
- Tangency: maximise $\mathbf{w}^\top \boldsymbol{\mu} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$; the condition $\boldsymbol{\mu} \propto \Sigma \mathbf{w}$ gives $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}$
- Maximum squared Sharpe ratio: $\theta^2 = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, the quantity estimated with bias in Section 3
- With Σ positive definite and $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, the normalised weights $\Sigma^{-1} \boldsymbol{\mu} / B$ have Sharpe ratio $\text{sign}(B) \theta$
 - ▶ Maximum only if $B > 0$; $B < 0$ gives the minimum, $B = 0$ no fully invested solution

Tobin: Separation



James Tobin, Nobel Prize in Economics 1981

Photo: unknown author (1962); public domain;
[Wikimedia Commons](#)

- Tobin (1958): add a riskless asset paying R_f
- Every investor holds the **same** risky portfolio, mixed with cash or borrowing
- Risk aversion changes only the mix, not the risky portfolio: **two-fund separation**
- The best risky portfolio maximises the Sharpe ratio: the **tangency portfolio**
- Separation is the bridge from Markowitz to the CAPM (Chapter 3)

Two Special Portfolios

Global minimum variance (GMV)

- Lowest variance of all portfolios

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

- Needs **no** expected returns

- In this chapter, MV denotes the tangency rule estimated from sample means and covariances

Tangency (maximum Sharpe)

- Highest Sharpe ratio

$$w_{\text{MV}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}$$

- μ : expected **excess** returns; requires $\mathbf{1}^\top \Sigma^{-1} \mu > 0$

Merton: the Frontier in Closed Form



Robert C. Merton, Nobel Prize in
Economics 1997

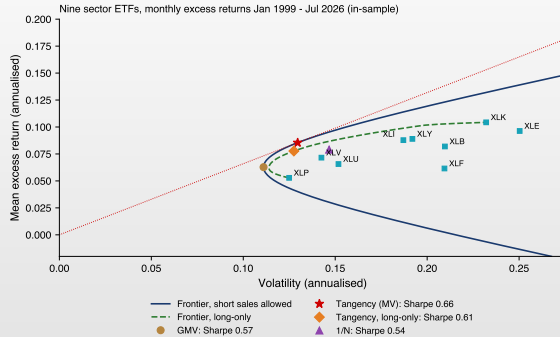
Photo: Digarnick (2006); public domain;
[Wikimedia Commons](#)

- Merton (1972): with short sales, the frontier is a hyperbola
- Σ positive definite, μ not proportional to 1 ; let $A = 1^\top \Sigma^{-1} 1$, $B = 1^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$, $D = AC - B^2 > 0$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- The GMV sits at $m = B/A$ with $\sigma^2 = 1/A$
- Every frontier portfolio mixes any two distinct frontier portfolios, e.g. GMV and tangency $\Sigma^{-1} \mu / B$ (if $B \neq 0$)

The Frontier of US Sectors



□ Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample

Reading the Sector Frontier

- ▣ Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- ▣ Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care
- ▣ Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero
 - ▶ A **corner solution**: the constraint binds and assets drop out
- ▣ The gain of the optimiser over $1/N$ is small even **in sample**, with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ [Frost & Savarino \(1988\)](#): constraining weights improved out-of-sample performance
- ▣ In this chapter: MV-LO and GMV-LO are the long-only versions of MV and GMV

Recap: Mean-Variance Optimisation

- ▣ The frontier trades mean against variance; correlation drives diversification
- ▣ Two special portfolios: GMV, which needs no means, and tangency, which maximises the Sharpe ratio
- ▣ With a riskless asset, the risky part of every efficient portfolio is the tangency portfolio
- ▣ Long-only constraints create corner solutions
- ▣ Everything so far assumed the parameters are **known**

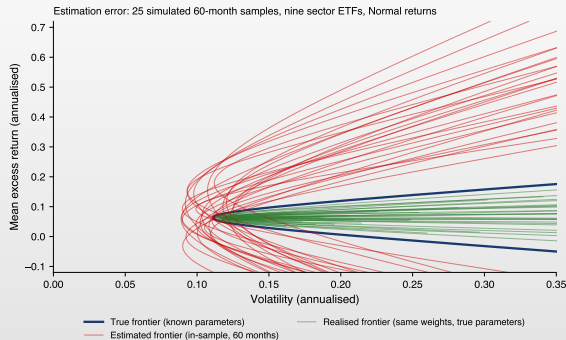
Means Are Hard to Estimate: Merton's Argument

- Merton (1980): Y years sampled h times a year, i.i.d. Normal returns (a diffusion) with annual mean μ and variance σ^2
 - ▶ $\text{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h ; only the **span** of the sample helps
 - ▶ $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$ as $h \rightarrow \infty$ (Normal; in general $(\kappa - 1)\sigma^4/(hY)$, κ the kurtosis): variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation

- Michaud (1989): mean–variance optimisers are **estimation-error maximisers**
 - ▶ They overweight assets with overestimated means and underestimated risk
- Best & Grauer (1991): tiny changes in a mean can move the weights of many assets
- In-sample performance is therefore **biased upwards**
- Experiment: simulate returns from known parameters, optimise, evaluate with the true parameters

Estimated versus Realised Frontiers



- True parameters: sector sample means and covariances, 1999–2026; returns drawn from the Normal distribution

Reading the Simulation

- ▣ Red: frontiers estimated from 60 months, far too optimistic
- ▣ Green: the same portfolios evaluated with the true parameters, all **below** the true frontier
- ▣ 2000 simulated tangency portfolios, 60 months each
 - ▶ Median estimated maximum Sharpe $\hat{\theta}$ 1.56; median true Sharpe 0.30; true optimum 0.66
 - ▶ In 11% of the samples $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$: the normalised weights are then the in-sample minimum
 - ▶ 96% have a true Sharpe below $1/N$ (0.54); 7.5% below zero
- ▣ Even with i.i.d. Normal returns and constant parameters, estimation error destroys the optimum

The Plug-in Sharpe Ratio Is Biased: an Exact Result

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood $\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}$: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart), independent
- Inverse Wishart moment, finite for $T > N + 2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Hence, with $\hat{\theta}^2 = \hat{\boldsymbol{\mu}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$ (Kan & Zhou (2007))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2} \left(\theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T-N-2}$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\mathbb{E}[\hat{\theta}^2]$ corresponds to a Sharpe of 1.65 a year
 - The analytical counterpart of the simulated median of 1.56 on the previous slide

Correcting the Bias and the Distribution of Weights

- Unbiased estimator of θ^2 (Kan & Zhou (2007)): $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N]/T$
 - ▶ It can be negative; truncating at zero gives a nonnegative but upward-biased estimator, and $\sqrt{\hat{\theta}_u^2}$ is not unbiased for θ
- Sectors, full sample ($T = 331$, $N = 9$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}_u^2} = 0.31$ a year
 - ▶ The correction removes more than half of the in-sample tangency Sharpe, even with 27 years of data
- GMV weights, i.i.d. Normal returns (Wishart $\hat{\Sigma}$) and T large enough relative to N : $\hat{w}_{\text{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $1^\top \hat{\Sigma}^{-1} \hat{\mu}$, which can be near zero: very heavy tails, the explosions of the rolling weights
- Estimation-risk-optimal mixtures of tangency, GMV and $1/N$: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Tangency Weights by Regression: Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- Sherman–Morrison on $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar 4, A2); $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$ sample tangency weights
- OLS t -test of $b_i = 0$: asset i has zero weight in the tangency portfolio
- F -test of $\mathbf{R}\mathbf{b} = 0$: for example $b_1 = \dots = b_N$, i.e. “the tangency portfolio is $1/N$ ”
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise

Britten-Jones on the Nine Sectors

- Monthly excess returns, January 1999 – July 2026 ($T = 331$); normalised $\hat{b}$ reproduces the tangency weights (-0.37 financials, $+0.37$ health care)
- No single weight differs significantly from zero: largest $|t| = 1.12$ (financials); with HAC errors 1.19
- H_0 : tangency = $1/N$, eight restrictions $b_i = b_9$
 - ▶ $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi^2_8 = 4.50$, $p = 0.81$
- 27 years of data cannot tell the estimated optimum from equal weights
- The same verdict as the window-length simulation and the out-of-sample tests below

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months (top panel clipped at ± 6)

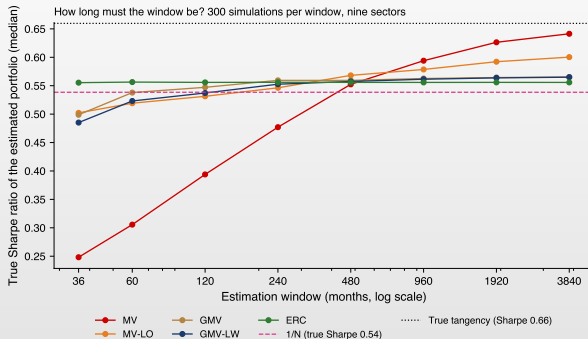
Reading the Rolling Weights

- ▣ MV: median gross exposure $\sum_i |w_i| = 4.9$; largest single weight 26 in absolute value
- ▣ Gross exposure above 3 in 90% of the months
- ▣ Explosions in 2004–2006 and 2009: $\mathbf{1}^\top \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$ close to zero in the denominator
 - ▶ Negative in 12 of 271 windows: MV then holds the minimum-Sharpe portfolio
- ▣ GMV with shrinkage (next section): gross exposure 1.5, largest weight 0.94
- ▣ Unstable weights mean high turnover, hence high trading costs

The 1/N Benchmark

- ▣ **1/N**: equal weights $w_i = 1/N$, rebalanced each month; no estimation at all
- ▣ DeMiguel, Garlappi & Uppal (2009): 14 optimisation models on 7 data sets
 - ▶ None beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
- ▣ Their calibration: about 3000 months of data for sample MV to beat 1/N with 25 assets
- ▣ Kan & Zhou (2007): combining tangency and GMV reduces estimation risk
- ▣ Question: how long must the window be for our nine sectors?

How Long Must the Window Be?



- 300 simulated samples per window length; true parameters as in the previous simulation

Reading the Window-Length Experiment

- ▣ MV beats 1/N only with 480 months (40 years) of data: median true Sharpe 0.55 vs 0.54
- ▣ With 60 months, MV reaches 0.31, less than 60% of 1/N
- ▣ Long-only MV, GMV and GMV with shrinkage are close to 1/N from 60 months on
- ▣ ERC (Section 6) is almost insensitive to the window: 0.56 throughout
- ▣ Here 1/N is nearly efficient: the true tangency Sharpe is only 0.66

Recap: The Limits of Markowitz

- ▣ Means are estimated with large errors; the optimiser amplifies them
- ▣ In-sample Sharpe ratios of optimised portfolios are biased upwards: $\mathbb{E}[\hat{\theta}^2] = (T\theta^2 + N)/(T - N - 2)$
- ▣ Britten-Jones regression: tangency weights are OLS coefficients normalised to sum to one; on sectors they are not statistically distinguishable from $1/N$
- ▣ Sample MV weights are unstable and extreme
- ▣ $1/N$ is a hard benchmark: MV needs decades of data to beat it
- ▣ Remedies: better covariance estimates, better means, or rules that avoid means

Why the Sample Covariance Matrix Fails

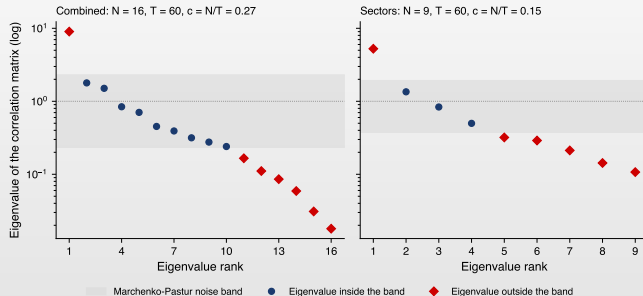
- ▣ N assets: $N(N + 1)/2$ variances and covariances; 16 assets: 136 parameters from 60 months
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$

Eigenvalues of Pure Noise: Marchenko–Pastur

- i.i.d. returns with $\Sigma = \sigma^2 I$: every true eigenvalue equals σ^2
- Marčenko & Pastur (1967): as $N, T \rightarrow \infty$ with $N/T \rightarrow c \in (0, 1)$, the sample eigenvalues fill
$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$
- $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
- Laloux et al. (1999): eigenvalues outside the band are candidates for structure beyond this noise model; those inside cannot be distinguished from it by this comparison alone
- This turns “sample eigenvalues are too dispersed” into a quantitative benchmark

Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); grey band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$

Reading the Eigenvalue Chart

- ▣ 16 ETFs: one eigenvalue above $\lambda_+ = 2.30$, the market mode (9.0, 56% of the trace); nine lie inside the band
- ▣ Nine sectors: again only the market mode (5.2) lies above $\lambda_+ = 1.92$
- ▣ Rescaling the noise variance to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)): three eigenvalues above the band in each universe
- ▣ Eigenvalues **below** the band (six of 16): near-riskless combinations, e.g. TLT against IEF
 - ▶ GMV loads exactly on these directions: if they are underestimated, its risk is too

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: [Ledoit & Wolf \(2003\)](#)
- ▶ Scaled identity: [Ledoit & Wolf \(2004a\)](#)
- ▶ Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: [Ledoit & Wolf \(2004b\)](#), used here

- δ trades the variance of S against the bias of F

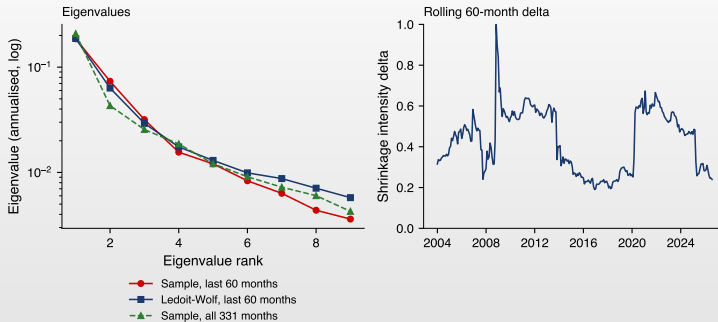
The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their covariances with the target; $\hat{\gamma} = \|F - S\|^2$ (misspecification)
- Noisy data or a good target: large δ ; long samples: $\delta \rightarrow 0$

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026

Reading the Shrinkage Chart

- ▣ Last window: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged
- ▣ Condition number: sample 51.6, shrunk 32.5
- ▣ Rolling δ : mean 0.43, range 0.19–1.00
 - ▶ $\delta = 1$ in the early windows: 60 months are too few to beat the target
- ▣ GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \lambda \geq 0 \text{ the multipliers of } w_i \geq 0$$

- ▶ $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers all its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk
- ▣ The effect is largest when covariances are estimated from few observations
- ▣ Practical rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

Beyond Linear Shrinkage

- Linear shrinkage mixes two matrices with one δ ; only towards a scaled identity does it just pull the eigenvalues to their mean, with the eigenvectors kept; **nonlinear** shrinkage keeps the sample eigenvectors and corrects each eigenvalue separately
 - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Factor structure plus a sparse residual covariance: POET (Fan, Liao & Mincheva (2013)), the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)
- Out of sample, 60-month window, GMV volatility
 - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
 - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix

Recap: Covariance Shrinkage

- The sample covariance matrix is noisy and ill-conditioned when N is large relative to T
- Shrinkage mixes it with a structured target; the data choose the weight δ
- The target matters: constant correlation suits one asset class better than a mix of stocks and bonds
- No-short constraints act as implicit shrinkage
- Marchenko–Pastur gives an asymptotic noise benchmark for the eigenvalues; nonlinear shrinkage treats each eigenvalue separately

Shrinking the Means: One Family of Estimators

- Bayes–Stein (Jorion (1986)): $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$
 - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and $1/N$ that minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints $\|w\|_1 \leq \delta$ or $\|w\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and $1/N$; the bridge to the lasso portfolios of Chapter 13
- Black–Litterman (next slides): Bayesian shrinkage of μ towards the returns implied by a benchmark

Black and Litterman: Starting from Equilibrium



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Black & Litterman (1992), developed at Goldman Sachs
- Problem: sample means give extreme, unintuitive weights
- Idea: start from the returns that make a **benchmark** portfolio optimal
- Add the investor's **views**, weighted by their confidence
- Result: weights that move away from the benchmark only where views exist

Step 1: Reverse Optimisation

- ▣ A mean–variance investor with risk aversion δ_{RA} and access to cash holds $w^* = (\delta_{RA} \Sigma)^{-1} \mu$ in risky assets and $1 - \mathbf{1}^\top w^*$ in cash
- ▣ Reverse it: the returns that make the benchmark w_b optimal

$$\pi = \delta_{RA} \Sigma w_b$$

- ▣ π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- ▣ Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{RA} = 2.5$

Step 2: Views and the Posterior

- ▣ K views: $P\mu = q + \varepsilon$, $\varepsilon \sim N(0, \Omega)$; prior $\mu \sim N(\pi, \tau\Sigma)$
- ▣ Posterior mean: a precision-weighted average

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \pi + P^\top \Omega^{-1} q]$$

- ▣ Weights: $w_{BL} = (\delta_{RA}\Sigma)^{-1} \mu_{BL}$
- ▣ **He & Litterman (2002)**: $\tau = 0.05$, $\Omega = \text{diag}(P \tau \Sigma P^\top)$; with Σ in the weights, only the assets in the views move (the predictive covariance $\Sigma + M$, next slide, would change all weights; without views it scales them by $1/(1 + \tau)$)

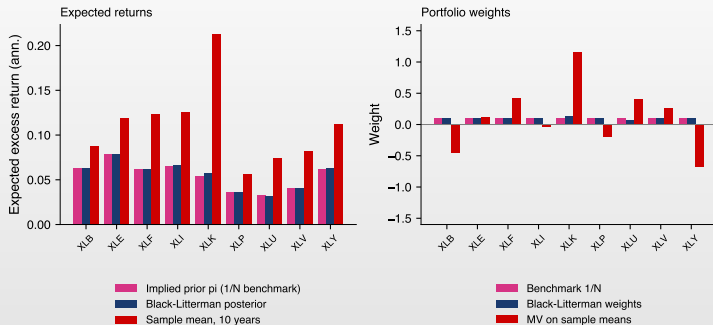
Black–Litterman as Mixed (GLS) Estimation

- ▣ Theil & Goldberger (1961): stack the prior and the views as one regression on μ

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- ▣ GLS gives exactly μ_{BL} of the previous slide, with covariance $M = [(\tau \Sigma)^{-1} + P^T \Omega^{-1} P]^{-1}$
- ▣ Equivalent form: $\mu_{BL} = \pi + \tau \Sigma P^T (P \tau \Sigma P^T + \Omega)^{-1} (q - P \pi)$
- ▣ Limits: $\Omega \rightarrow 0$ gives $P \mu_{BL} = q$ exactly (restricted least squares); $\Omega \rightarrow \infty$ gives π
- ▣ One asset, $\pi = 5\%$, $\tau \sigma^2 = 0.00128$, view 8% with variance 0.0004: $\mu_{BL} = 7.29\%$, view weight 0.76

Black-Litterman on US Sectors



□ Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year

Reading Black–Litterman

- ▣ Implied XLK minus XLU: 2.2%; view 3%; posterior 2.6% (the average, since Ω equals the prior variance of the view)
- ▣ Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged
- ▣ Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- ▣ The view is relative ($P1 = 0$): the tilt $\propto P^\top$ sums to zero, so the portfolio stays fully invested; an absolute view would also move the cash position
- ▣ Black–Litterman is Bayesian shrinkage of the means towards equilibrium

Recap: Black–Litterman

- The prior comes from a benchmark, not from noisy sample means
- Views enter with explicit confidence; weights move only where views exist
- Results depend on the benchmark, τ , Ω and the risk aversion
- Without views, Black–Litterman returns the benchmark itself
- Formally a GLS (mixed) estimator: one member of the family of mean-shrinkage estimators with Bayes–Stein and 1/N combinations

Risk Contributions

- ▣ Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- ▣ **Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share RC_i / σ_p
- ▣ Equal weights are not equal risks: volatile assets dominate
- ▣ Risk budgeting: choose w so that the shares RC_i / σ_p equal preset budgets b_i

Equal Risk Contribution

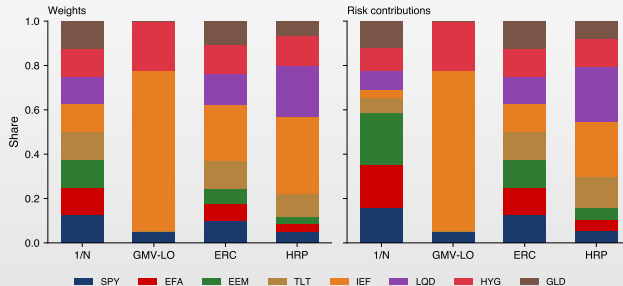
- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j
- Computed from a convex problem, then normalised to sum to one

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
 - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY; a 60/40 mix lets SPY carry 94% of the risk
- Needs no expected returns, only Σ ; for Σ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)

Reading the Risk Contributions

- ▣ 1/N: equities and high yield receive half of the weight but 69% of the risk
- ▣ ERC: 12.5% of the risk from every asset; IEF gets the largest weight (25%)
- ▣ Long-only GMV: 73% IEF, 22% HYG, 5% SPY; five assets at zero
- ▣ Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%
- ▣ ERC lies between 1/N and GMV, as the theory predicts

Hierarchical Clustering of Assets

- ▣ Correlation distance: $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$
- ▣ **Single linkage**: repeatedly merge the two closest clusters; cluster distance = closest pair
- ▣ The result is a tree (**dendrogram**); cutting it gives groups of similar assets
- ▣ Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- ▣ Textbook: Härdle & Simar (2019), cluster analysis

Hierarchical Risk Parity

▣ HRP (López de Prado (2016)), three steps

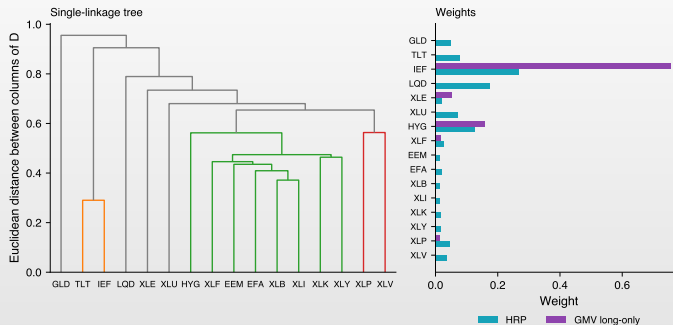
- ▶ Tree clustering: Euclidean distance between the columns of $D = (d_{ij})$, $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$, then single linkage on $\tilde{d}_{ij}$
- ▶ Quasi-diagonalisation: reorder assets so that similar ones are adjacent
- ▶ Recursive bisection: split the ordered list in two; split the weight in inverse proportion to the variance of each half

▣ Variance of a half: inverse-variance weights inside it, $\tilde{w}^\top \Sigma \tilde{w}$

▣ No matrix inversion: works even when Σ is singular

▣ All weights are positive

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026

Reading the Tree

- ▣ Closest pair: TLT and IEF ($\tilde{d} = 0.29$); gold joins last (0.96)
- ▣ Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
- ▣ Defensive sectors (utilities, staples, health care) and energy join the equity cluster late
- ▣ HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6% and HYG 12.6%
- ▣ Long-only GMV: 75.6% in IEF and only five assets above zero

Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions
 - ▶ ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Recap: Risk-Based Allocation

- ▣ Risk contributions decompose portfolio volatility additively
- ▣ ERC equalises them; it lies between $1/N$ and GMV
- ▣ HRP uses a correlation tree and needs no matrix inversion
- ▣ These rules avoid expected returns entirely: less estimation error, no view on returns

Backtest Design

- ▣ **Out-of-sample** (OOS): weights at month t use only data up to $t - 1$
- ▣ Rolling window of 60 months; rebalancing every month
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes: sectors (OOS January 2004 – July 2026, 271 months), multi-asset and combined (May 2012 – July 2026, 171 months)
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth

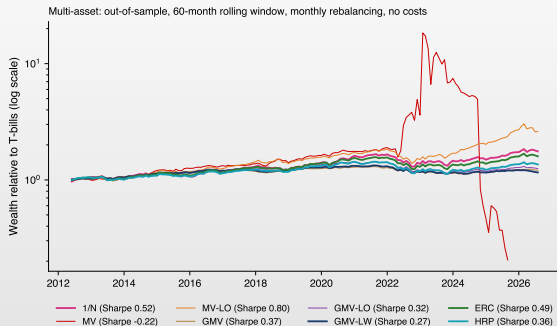
Turnover and Transaction Costs

- Between rebalancings weights drift with returns

$$w_{i,t}^+ = \frac{w_{i,t}(1 + R_{i,t})}{\sum_j w_{j,t}(1 + R_{j,t})}$$

- Turnover:** traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$
- 1/N also trades: it sells winners and buys losers every month

Out-of-Sample Growth: Multi-Asset

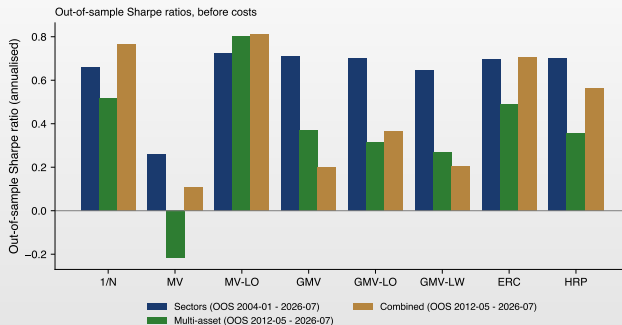


- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs; a path ends at a monthly loss of 100% or more (MV)

Reading the Multi-Asset Backtest

- ▣ MV collapses: annualised volatility 518%, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- ▣ MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- ▣ GMV rules have the lowest volatility (3.8%–5.4%) but also low Sharpe ratios (0.27–0.37)
 - ▶ They load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5% , IEF -0.2% a year)
- ▣ Is MV-LO really better, or lucky? A test is needed

Sharpe Ratios Across Universes



60-month rolling window, monthly rebalancing, annualised, before costs

MFM_ch4_oos_backtest

Results: Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

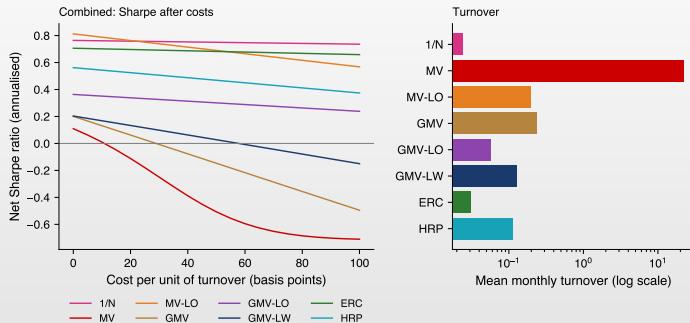
- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series, not investable wealth
- All rules except MV land within 0.07 of 1/N

Results: Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover
- 1 basis point (bp) = 0.01%

Sharpe Ratios after Costs



Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover

Reading the Cost Chart

- ▣ Turnover ranges from 0.02 (1/N) to 22 (MV) per month
- ▣ Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
- ▣ MV-LO overtakes 1/N only for costs below about 22 bp (Seminar, B8)
- ▣ 1/N and ERC barely move: low turnover is a hidden advantage
- ▣ Rule: report net performance for realistic costs, not only gross

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$, with SR monthly
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
 - ▶ Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$) is exact only without autocorrelation; in general $SR(q) = \eta(q) SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}}$$

- Here $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$: $\eta = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe 0.69

Testing a Difference in Sharpe Ratios

- ▣ H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months
- ▣ Jobson & Korkie (1981): a test under i.i.d. Normal returns
- ▣ Ledoit & Wolf (2008): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- ▣ Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap



Bradley Efron, inventor of the bootstrap
(1979)

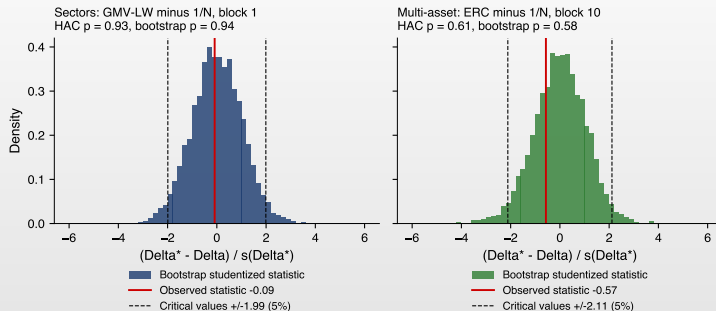
Photo: Ryan K. Morris (2007); public
domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of pairs (r_{1t}, r_{2t}) , wrapping around the end of the sample
- **Studentized**: bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$, with $s(\hat{\Delta}^*)$ from the block sums of each resample
 - ▶ The studentized statistic is asymptotically pivotal: more accurate than the Normal approximation of the HAC test
- $s(\hat{\Delta})$ on the data: prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)), automatic bandwidth (Andrews (1991))
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$, $M = 4999$ draws

Choosing the Block Length: Algorithm 3.1

- Ledoit & Wolf (2008), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
 - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
 - ▶ Generate $K = 5000$ pseudo-samples of length T ; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
 - ▶ $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- Chosen blocks: mostly 1 for sectors, 8–10 for multi-asset and combined rules (stronger dependence)
- A non-studentized bootstrap with a fixed 6-month block is not this test: its block is not calibrated for coverage

Is Anything Better than $1/N$?



- Bootstrap distributions of the studentized Sharpe ratio difference against $1/N$; red: observed statistic; dashed: bootstrap critical values

Reading the Tests

- ▣ Sectors, GMV-LW minus $1/N$: -0.02 , 95% interval $[-0.36, 0.33]$; HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1)
- ▣ Multi-asset, ERC minus $1/N$: -0.03 , interval $[-0.15, 0.08]$; HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10)
 - ▶ ERC and $1/N$ are so correlated that the interval is narrow, yet it still covers zero
- ▣ Sectors: no rule differs from $1/N$ at 5%; the closest is ERC, $+0.04$ (HAC $p = 0.09$, bootstrap 0.13)
- ▣ Multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

Significant Differences: Mostly Losses

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.57	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.82

- No rule beats 1/N significantly in any universe
- Unadjusted, GMV rules lose to 1/N on the combined universe (HAC $p \leq 0.05$, bootstrap p from 0.03 to 0.08): low risk, but too little return
- 21 tests: Holm-adjusted p -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives (Chapter 3, multiple testing)

Many Rules, One Winner: the Deflated Sharpe Ratio

- ▣ Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- ▣ Expected maximum under H_0 : all true Sharpe ratios zero, N independent trials, monthly units as T ; with $\gamma_E = 0.5772$ (Euler's constant) and $\hat{V}$ the variance of the N estimates

$$SR_0 = \sqrt{\hat{V}} \left[(1 - \gamma_E) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- ▣ Deflated Sharpe ratio, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$DSR = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
 - ▶ As 24 independent trials: $SR_0 = 0.52$ a year, $DSR = 0.84 < 0.95$
 - ▶ The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98; the verdict hinges on the effective N

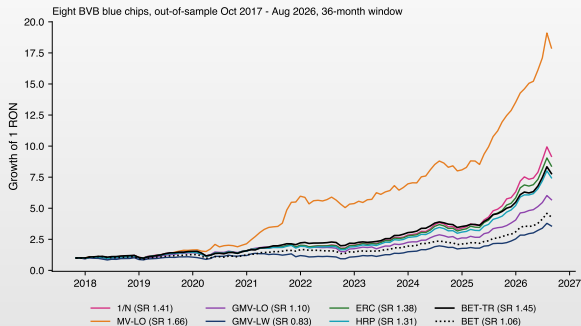
Recap: Out-of-Sample Evaluation

- ▣ Only out-of-sample, net-of-cost results are informative
- ▣ Sample MV fails in every universe; constraints and shrinkage tame it
- ▣ Risk-based rules and $1/N$ are hard to separate statistically
- ▣ Test Sharpe differences with HAC errors or the studentized block bootstrap, and count your tests
- ▣ Single Sharpe ratios need standard errors and autocorrelation-aware annualisation; a winner among many needs deflation

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



□ Out-of-sample, October 2017 – August 2026 (107 months), before costs

Reading the BVB Backtest

- ▣ BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07
 - ▶ The gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- ▣ 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- ▣ MV-LO 1.66 (difference +0.21, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- ▣ Average correlation of the eight stocks: 0.44; the latest GMV-LO puts 70% in TLV

Beyond Equities: Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#); Seminar 4, B5

Recap: BVB Portfolios

- Benchmark choice matters: BET omits about 7% a year of dividends
- No optimised BVB-only rule beats BET-TR significantly
- Eight stocks with average correlation 0.44: little room for diversification inside the BVB
- Open question: does diversification **beyond** BET pay? (Seminar 4, Part C)

Case Study: Which Frontier Can a Large Investor Reach?

- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*: how should portfolios be constructed from machine-learning return forecasts when every trade affects prices?
- The rules of this chapter fight estimation risk; Ledoit & Wolf (2017) is a key reference for estimating the covariance matrix $\hat{\Sigma}$, but costs enter at most as a flat rate per unit of turnover
- Implementable efficient frontier (eq. 12): for each volatility σ , the highest expected return **net of trading costs**
 - ▶ $k(\sigma) = \max_{\pi} E[r_{t+1}^{\text{net}}]$ subject to $E[\pi_t^{\top} \Sigma \pi_t] = \sigma^2$
 - ▶ Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier, and a larger investor faces a lower frontier
- Usual two-step practice: forecast returns with ML, then optimise; the forecasts load on short-lived signals of small stocks, whose turnover eats the return
- The paper's method generalises the dynamic trading rule of Gârleanu & Pedersen (2013) and learns the weights directly from the signals, as Brandt, Santa-Clara & Valkanov (2009) do without costs

Case Study: Data and Method

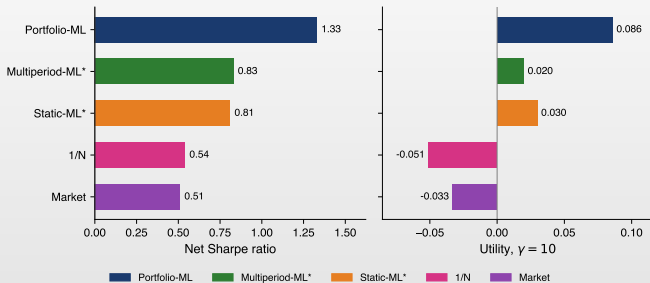
- Trading cost as a fraction of wealth W_t (eqs. 4, 7): $c_t = \frac{W_t}{2} (\pi_t - g_t \pi_{t-1})^\top \Lambda (\pi_t - g_t \pi_{t-1})$
 - ▶ π_t : risky-asset weights, $1 - 1^\top \pi_t$ in the risk-free asset; $g_t = \text{diag}((1 + r_t^f + r_t)/(1 + g_t^w))$ (eq. 6): drift of last month's weights, g_t^w the growth of wealth; $\Lambda = \text{diag}(0.2/V_i)$, V_i the average daily dollar volume over 6 months: trading 1% of V_i moves the price by 0.1% (eq. 35, [Frazzini, Israel & Moskowitz \(2018\)](#))
- Optimal rule (Proposition 3): $\pi_t = m g_t \pi_{t-1} + (1 - m) A_t$: trade part of the way towards an aim portfolio A_t , a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14)
- Portfolio-ML (Proposition 4, eq. 40): $A_t = \text{diag}(1/\sigma_{i,t}) f(s_t) \beta$, $f = p$ random Fourier features of the signals s_t
 - ▶ $\hat{\beta} = (E_T[\tilde{\Sigma}_t] + \lambda I)^{-1} E_T[\tilde{r}_{t+1}]$: a ridge regression that maximises utility net of costs, in closed form (eqs. 24–26)
- Data (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks), 115 characteristics of [Jensen, Kelly & Pedersen \(2023\)](#) ranked into $[0, 1]$; December 1952 – December 2020, out of sample 1981–2020, monthly
- Σ : 13 characteristic themes and 12 industries as factors, exponential weights with half-lives of 378 days (correlations) and 126 days (variances); investor: $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market (g_t^w = market return; flows in and out keep it so, Section 5.1.3)
- Tuning (Table 1): $\lambda \in \{0, e^{-10}, e^{-9.8}, \dots, e^{10}\}$, $p \in \{2^6, \dots, 2^9\}$, $\eta \in \{e^{-3}, e^{-2}\}$ (standard deviation of the random weights), chosen each year by the utility of validation backtests since 1971
- Rivals: Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22), factor sorts, $1/N$, minimum variance, market; “*” marks an extra tuning layer (Section 5.3)

Case Study: Key Results

- Table 2, Markowitz-ML: gross Sharpe ratio 2.06, but monthly turnover 79.36 and leverage 74.61: net return extremely negative
 - ▶ The textbook frontier exists only on paper
- Table 2, Portfolio-ML: net Sharpe ratio 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year, turnover 0.25)
 - ▶ Utility (eq. 11): annualised net return minus $\frac{\gamma}{2}$ times the annualised variance
- Table 2, rivals (net Sharpe ratio, utility): Static-ML* 0.81, 0.030; Multiperiod-ML* 0.83, 0.020; 1/N 0.54, -0.051; market 0.51, -0.033
 - ▶ Only cost-aware rules deliver positive utility; Portfolio-ML needs no second tuning layer
- Table 3: posterior probability that Portfolio-ML has the higher utility: 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- Figure 1(B) (Section 6.2): with \$10 billion and $\gamma = 10$, net excess return 14.6% at 11% volatility; with \$1 billion, 16.3% at the same volatility
 - ▶ The frontier depends on who invests: size is a cost
- Figure 5: net of costs, quality is the most important theme and short-term reversal almost irrelevant; monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3)
 - ▶ Persistent signals are cheap to trade

Case Study: Table 2 at a Glance

Out of sample 1981-2020, US stocks, wealth \$10 billion (Markowitz-ML off scale)

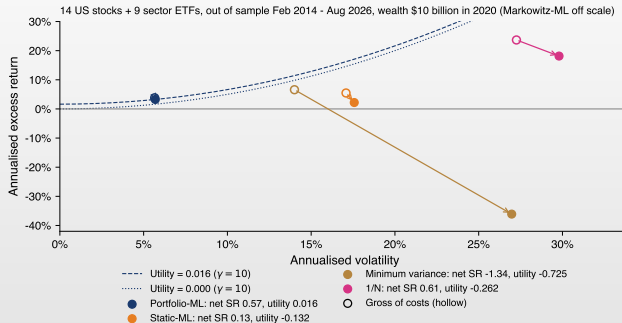


- Source: Jensen, Kelly, Malamud & Pedersen (2026), Table 2; net of trading costs, out of sample 1981–2020, $\gamma = 10$
- Utility and net Sharpe ratio rank 1/N and the market differently: the variance penalty on the extra volatility of 1/N exceeds its higher return

Case Study: Replication, Limits and an Open Extension

- ▣ Course data: 14 US stocks and 9 sector ETFs (data/market); five signals from prices and volumes (1-month return, momentum from $t - 6$ and from $t - 12$ to $t - 1$, 1-year volatility, 6-month dollar volume); same cost rule, γ , wealth and grid; Σ with the same weights on the 23 assets; validation from 2011, out-of-sample returns February 2014 – August 2026 (151 months)
- ▣ Portfolio-ML: net Sharpe ratio 0.57, utility 0.016, cost 0.6% a year, leverage 0.32; 1/N: net Sharpe ratio 0.61, but utility -0.26 (net volatility 30%)
 - ▶ Utility: Static-ML -0.13 ; minimum variance -0.72 (cost 43% a year); Markowitz-ML extremely negative (leverage 18)
 - ▶ Posterior probability of a higher utility for Portfolio-ML: 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- ▣ Same ranking of cost-aware against cost-agnostic rules as in Table 2; Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- ▣ Limits of the course data: 23 assets instead of 184–805, five signals instead of 115, no factor model for Σ , whose window reaches 10 years only in August 2014 (the common-price panel starts in August 2004)
Deviations of this run: return forecasts with $p \geq 2^6$ and three validation years (paper: 2^1 – 2^9 , ten years), no starred rivals
- ▣ Limits stated by the authors: fees and other real-world frictions, computing power and ML methods not available in real time early in the sample (Section 6.1); re-estimated each month, but costs and risk treated as constant over the horizon of each optimisation (Section 5.1.3); among the smallest 20% of stocks Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)
- ▣ Open extension: implementable frontiers on the course data for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$: where does 1/N, costed at each wealth, fall relative to them? (Seminar 4, C3)

Case Study: Gross and Net of Costs on the Course Data



- Out of sample, February 2014 – August 2026 (151 months); hollow marker: before costs; filled marker: after the costs of eq. (35)

Case Study: Reading the Replication

- ▣ Dashed curve: all volatility–return pairs with the utility of Portfolio-ML (0.016); dotted curve: utility 0
 - ▶ Any point below the dashed curve is worse for an investor with $\gamma = 10$
- ▣ Portfolio-ML barely moves: 3.8% gross, 3.3% net a year, at 5.7% volatility
- ▣ Minimum variance falls from +6.6% gross to −36.1% net: rebalancing a leverage of 2.6 costs 43% a year
- ▣ 1/N keeps 18.1% net, but at 29.8% volatility it lies far below the dashed curve
- ▣ The frontier that matters is the one after costs ([Jensen, Kelly, Malamud & Pedersen \(2026\)](#), eq. 12)

An Open Question

- Does any **estimated** allocation rule beat $1/N$ out of sample, net of costs, once every rule, window and universe tried is counted?
- Live evidence points both ways
 - ▶ Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
 - ▶ Combining tangency, GMV and $1/N$ under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
 - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for K rules on one universe
 - ▶ H_0 : $\Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; H_1 : $\Delta_k > 0$ for some k
 - ▶ Statistic: $\hat{\Delta}_k$ with HAC or block-bootstrap errors ([Ledoit & Wolf \(2008\)](#)); error rate fixed for the whole family
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with $1/N$ out of sample; for each give DOI, N , T , cost assumption and test."*
 - ▶ Resolve every DOI on Crossref and open the table cited; assistants merge authors and invent DOIs
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over $1/N$ (better Σ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads data/market, joins prices on common dates and runs a 60-month rolling GMV-LW and $1/N$; weights at t may use data up to $t - 1$ only."*
 - ▶ Replicate first: the combined-universe table of this chapter ($1/N$ 0.76, GMV-LW 0.20) before changing anything

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests ($\text{rules} \times \text{universes} \times \text{windows} \times \text{costs}$), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
 - ▶ Placebo: random long-only weights with the same turnover; a real effect must beat their distribution
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted p -values, the power of the design, code and seeds
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Look-ahead:** weights at t built with month- t returns $\Rightarrow$ assert that the last estimation date is before t ; lagging the weights one more month is only a sensitivity check
- **Survivorship and selection:** an ETF list chosen today $\Rightarrow$ use the universe available at each date, including later delisted ETFs and their final returns; otherwise disclose the limitation
- **Multiple testing:** 7 rules $\times$ 3 universes already give 21 tests $\Rightarrow$ Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted p -values
- **Power:** $\hat{\Delta}$ has a standard error near 0.18 on 171 months $\Rightarrow$ compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns; Sharpe ratios use **excess** returns $\Rightarrow$ recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch $\Rightarrow$ drop the claim, log it in `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest

- A typical assistant proposal: “MV-LO beats 1/N on the multi-asset universe”; $H_0: \Delta \leq 0$, one-sided $p = 1 - \Phi(\hat{\Delta}/\text{se})$
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16, Section 7)
- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
 - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC p -values below 0.05 are all **losses**, which never reject a superiority null
 - ▶ One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

Your Turn: A Team-Project Seed

- Question: does nonlinear shrinkage ([Ledoit & Wolf \(2017\)](#)) or a three-fund combination ([Kan, Wang & Zhou \(2022\)](#)) beat $1/N$ on the 16 ETFs of this chapter, after 10–50 bp costs?
- Replicate first: the combined-universe table (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted p -values of the mini-case
- Deliverables
 - ▶ A one-page pre-registration: hypotheses, family of tests, error rate, power, placebo
 - ▶ Code on data/market, a table of all specifications, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (invented DOI, look-ahead, wrong return type) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_04/ai_discovery_case.py](#)

Conclusions

- ▣ Markowitz is the right question with fragile answers: estimation error dominates
- ▣ Sample MV needs decades of data to beat $1/N$; out of sample it fails in every universe
- ▣ Shrinkage, constraints, Black–Litterman and risk-based rules make portfolios stable
- ▣ No rule beats $1/N$ significantly here; several lose after costs
- ▣ Evaluate out of sample, net of costs, with a test of the Sharpe difference

Links to Other Chapters

- ▣ **Chapter 3** (factor models): the tangency portfolio and the CAPM; factor covariance models
- ▣ **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- ▣ **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- ▣ **Chapter 8**: backtesting without overfitting
- ▣ **Chapter 13**: machine learning for returns and portfolios

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ▣ Risk-free rate: Kenneth French Data Library

References (1/10)

- ▣ Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- ▣ Andrews, D.W.K., Monahan, J.C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, 60(4), 953–966.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- ▣ Best, M.J., Grauer, R.R. (1991). On the sensitivity of mean-variance-efficient portfolios to changes in asset means: some analytical and computational results. *Review of Financial Studies*, 4(2), 315–342.
- ▣ Black, F., Litterman, R. (1992). Global portfolio optimization. *Financial Analysts Journal*, 48(5), 28–43.

References (2/10)

- Bodnar, T., Parolya, N., Schmid, W. (2018). Estimation of the global minimum variance portfolio in high dimensions. *European Journal of Operational Research*, 266(1), 371–390.
- Brandt, M.W., Santa-Clara, P., Valkanov, R. (2009). Parametric portfolio policies: exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies*, 22(9), 3411–3447.
- Britten-Jones, M. (1999). The sampling error in estimates of mean-variance efficient portfolio weights. *Journal of Finance*, 54(2), 655–671.
- Chopra, V.K., Ziemba, W.T. (1993). The effect of errors in means, variances, and covariances on optimal portfolio choice. *Journal of Portfolio Management*, 19(2), 6–11.
- Choueifat, Y., Coignard, Y. (2008). Toward maximum diversification. *Journal of Portfolio Management*, 35(1), 40–51.
- DeMiguel, V., Garlappi, L., Nogales, F.J., Uppal, R. (2009). A generalized approach to portfolio optimization: improving performance by constraining portfolio norms. *Management Science*, 55(5), 798–812.

References (3/10)

- ▣ DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- ▣ Dragotă, M., Șerbănescu, C.I., Pele, D.T. (2008). Portfolio diversification and market share analysis for Romanian insurance companies. In *Risk Management and Value*, World Scientific, 277–298.
- ▣ Fan, J., Liao, Y., Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B*, 75(4), 603–680.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Israel, R., Moskowitz, T.J. (2018). Trading costs. SSRN Working Paper 3229719.
- ▣ Frost, P.A., Savarino, J.E. (1988). For better performance: constrain portfolio weights. *Journal of Portfolio Management*, 15(1), 29–34.

References (4/10)

- ▣ Gârleanu, N., Pedersen, L.H. (2013). [Dynamic trading with predictable returns and transaction costs](#). *Journal of Finance*, 68(6), 2309–2340.
- ▣ Härdle, W.K., Simar, L. (2019). [Applied Multivariate Statistical Analysis](#), 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y. (2015). [Backtesting](#). *Journal of Portfolio Management*, 42(1), 13–28.
- ▣ He, G., Litterman, R. (2002). [The intuition behind Black-Litterman model portfolios](#). SSRN Working Paper 334304.
- ▣ Jagannathan, R., Ma, T. (2003). [Risk reduction in large portfolios: why imposing the wrong constraints helps](#). *Journal of Finance*, 58(4), 1651–1683.
- ▣ Jensen, T.I., Kelly, B., Malamud, S., Pedersen, L.H. (2026). [Machine learning and the implementable efficient frontier](#). *Review of Financial Studies*, 39(10), 3035–3078.

References (5/10)

- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). *Is there a replication crisis in finance?*. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jobson, J.D., Korkie, B.M. (1981). *Performance hypothesis testing with the Sharpe and Treynor measures*. *Journal of Finance*, 36(4), 889–908.
- ▣ Jorion, P. (1986). *Bayes-Stein estimation for portfolio analysis*. *Journal of Financial and Quantitative Analysis*, 21(3), 279–292.
- ▣ Kan, R., Wang, X., Zhou, G. (2022). *Optimal portfolio choice with estimation risk: no risk-free asset case*. *Management Science*, 68(3), 2047–2068.
- ▣ Kan, R., Zhou, G. (2007). *Optimal portfolio choice with parameter uncertainty*. *Journal of Financial and Quantitative Analysis*, 42(3), 621–656.
- ▣ Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.

References (6/10)

- ▣ Laloux, L., Cizeau, P., Bouchaud, J.-P., Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470.
- ▣ Ledoit, O., Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5), 603–621.
- ▣ Ledoit, O., Wolf, M. (2004a). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- ▣ Ledoit, O., Wolf, M. (2004b). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- ▣ Ledoit, O., Wolf, M. (2017). Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks. *Review of Financial Studies*, 30(12), 4349–4388.

References (7/10)

- ▣ Ledoit, O., Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5), 3043–3065.
- ▣ Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- ▣ López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4), 59–69.
- ▣ Maillard, S., Roncalli, T., Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- ▣ Mantegna, R.N. (1999). Hierarchical structure in financial markets. *European Physical Journal B*, 11(1), 193–197.
- ▣ Marčenko, V.A., Pastur, L.A. (1967). Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4), 457–483.

References (8/10)

- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ Merton, R.C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4), 1851–1872.
- ▣ Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- ▣ Michaud, R.O. (1989). The Markowitz optimization enigma: is 'optimized' optimal?. *Financial Analysts Journal*, 45(1), 31–42.
- ▣ Okhrin, Y., Schmid, W. (2006). Distributional properties of portfolio weights. *Journal of Econometrics*, 134(1), 235–256.
- ▣ Pele, D.T. (2023). Robust Markowitz for crypto-assets. Quantinar course, Bucharest University of Economic Studies.

References (9/10)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?*. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Petukhina, A., Trimborn, S., Härdle, W.K., Elendner, H. (2021). *Investing with cryptocurrencies – evaluating their potential for portfolio allocation strategies*. *Quantitative Finance*, 21(11), 1825–1853.
- ▣ Politis, D.N., Romano, J.P. (1994). *The stationary bootstrap*. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Sharpe, W.F. (1966). *Mutual fund performance*. *Journal of Business*, 39(S1), 119–138.
- ▣ Theil, H., Goldberger, A.S. (1961). *On pure and mixed statistical estimation in economics*. *International Economic Review*, 2(1), 65–78.
- ▣ Tobin, J. (1958). *Liquidity preference as behavior towards risk*. *Review of Economic Studies*, 25(2), 65–86.

References (10/10)

- Tu, J., Zhou, G. (2011). Markowitz meets Talmud: a combination of sophisticated and naive diversification strategies. *Journal of Financial Economics*, 99(1), 204–215.
- Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.